

EARTHPRESS · REQUIREMENTS

EarthPress requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	EPR-REQ-001	Version	0.3
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/earthpress		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	Status of every requirement from EPR-CAL-001; reference block and 2 MPa target adopted for TRL 3 (EPR-DDR-001); proposed revisions to R5, R7 and R13 listed, awaiting Amish
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

EarthPress requirements

These requirements are checked by calculation in EPR-CAL-001 (TRL 3). Targets are unchanged from v0.2: none of the TRL 2 recommendations adopted in EPR-DDR-001 relaxed or redefined a target, and the revisions proposed at TRL 3 (listed below) await Amish. Targets still need validation with users in co-design sessions (see EPR-PRB-001).

The **reference block** is 290 x 140 x 90 mm, about 3.65 L and 6.94 kg dry, from a sandy clay loam stabilized with 5 % cement by dry mass (block size, 2 MPa target and stabilizer adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review; EPR-DDR-001 items 3 to 5).

Table 1. Requirements and status at TRL 3 (EPR-CAL-001).

ID	Requirement	Target	Verification (TRL 3 or later)	Status (EPR-CAL-001)
R1	Block dimensions	290 x 140 mm $\pm$ 2 mm on plan; height 90 mm $\pm$ 3 mm, checked with the block gauge	Mold tolerance analysis; later gauge checks on a production run	Met on paper: 90.0 mm at the crank stop when the fill is within 0 to +2.7 %; mold bulge 0.08 mm
R2	Compaction pressure	2.0 MPa or more on the block face at the end of the stroke	Linkage force calculation	At risk: 81.2 kN reached with two operators (peak 668 N at the grip) and a weighed fill; one operator at 500 N stalls at 0.53 MPa
R3	Block strength with suitable soil	28-day dry compressive strength 4 MPa or more and wet strength 2 MPa or more (Auroville class B); in all cases 2.1 MPa (300 psi) or more, as in 14.7.4 NMAC	Literature review at TRL 3; block tests at TRL 4	Not verifiable at TRL 3
R4	Output	300 or more blocks in an 8 h day with a crew of four, including mixing	Cycle time estimate; later timed trials	At risk: 296 blocks a day at an assumed 85 s cycle
R5	Operator force	Peak pull of 500 N or less at the lever grip; grip between 0.9 and 1.6 m above ground through the stroke	Force calculation; ergonomic check of the lever arc	Not met: peak 668 N (334 N each with two operators); grip 0.86 to 1.89 m. The band allows only 24.5° of arc with a 1.65 m grip radius. Eject 468 N
R6	Garage-buildable	Built with a stick welder, angle grinder, drill press and hand tools; no turned or milled parts except purchased pins and bushings	Design review of every part	Met (design review): turned pins, shafts and bushes are bought
R7	Movable	Complete press 140 kg or less; no single piece over 50 kg after removing pins and bolts; fits a 1.5 m pickup bed	Mass estimate from the model	Not met: 187 kg in total; heaviest piece 36.6 kg; plan 1.28 x 0.58 m
R8	Durability	50,000 blocks before mold liners or bushings need replacing; all pins in replaceable bushings; grease points on every pivot	Wear estimate; later endurance log	Not verifiable at TRL 3: fatigue range 89 MPa against 243 MPa allowed; wear needs soil data
R9	Soil test kit	Field tests (jar sedimentation, shrinkage box, ribbon and drop tests) give a go or no-go and a starting cement content, with a first screen within 1 h and a full result within 24 h	Written procedure; later comparison with lab grading on partner soils	Met (design review); accuracy not verifiable at TRL 3
R10	Low stabilizer use	Blocks reach R3 with 8 % cement by dry mass or less; 5 % as the design target	Literature at TRL 3; block tests at TRL 4	Not verifiable at TRL 3 (design at 5 %, 0.33 kg per block)
R11	Embodied carbon	Cradle-to-gate CO ₂ of the walling material 25 % or less of a fired-brick wall of equal volume	Carbon estimate from published factors	Met on paper: 7.6 % (0.35 t against 4.55 t for a small house)
R12	Safe operation	Lever cannot fall freely when released under load; lid latch holds against at least	Design review; later proof-load test	Met on paper: rest catch and end pawl; lid and latch checked

ID	Requirement	Target	Verification (TRL 3 or later)	Status (EPR-CAL-001)
		1.5 times the peak block force; pinch points between lever and frame guarded or with 100 mm or more clearance		at 173 kN (1.5 times is 121.8 kN); linkage guard
R13	Affordable	Press and test kit \$450 or less in parts, excluding labor and cement	Priced BOM (bom/ bom.csv)	Not met: \$488 (indicative)
R14	Open and documented	CERN-OHL-S drawings, BOM and build notes sufficient for a welder to build without contacting the author	Review by an outside fabricator at TRL 3	Not verifiable at TRL 3: no outside review yet

■ Requirements not met or at risk

- **R5 not met:** one operator would need 668 N at the peak, and the 0.9 to 1.6 m band cannot hold a useful arc. Proposed rewording in EPR-DDR-001 item 12.
- **R7 not met:** 187 kg against 140 kg; every piece is under 37 kg. Proposed relaxation to 190 kg in EPR-DDR-001 item 13.
- **R13 not met:** \$488 against \$450. Recommended budget \$500, awaiting Amish (EPR-DDR-001 item 14).
- **R2 at risk:** met only with two operators, a weighed fill (0 to +2.7 %) and the assumed soil stiffness; a softer soil (20 mm per decade) raises the peak to 809 N.
- **R4 at risk:** 296 against 300 blocks a day on an assumed cycle.
- **R3, R8, R10 and R14 cannot be verified at TRL 3.**

■ Proposed revisions, awaiting Amish

These are not applied; the targets above are unchanged.

- R5: 500 N or less per operator; grip between 0.8 and 1.9 m, with the peak force at 1.0 m or higher.
- R7: complete press 190 kg or less; the 50 kg piece limit and pickup bed unchanged.
- R13: \$500 or less, if Amish raises `budget_usd`.

■ Assumptions

- Compaction pressure of 2 to 4 MPa and about 5 % cement follow Auroville Earth Institute practice ([Auroville Earth Institute](#)).
- Dry block density about 1,900 kg/m³ (estimate), giving about 6.9 kg per dry block.
- One operator can pull 500 N at the grip for short strokes, using body weight; two can pull 1,000 N on the T-handle. To be checked with users.
- Soil pressure rises tenfold over the last 14 mm of the stroke (range 10 to 20 mm); to be measured with partner soils.
- A crew of four covers soil sieving, mixing, pressing and stacking.